

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-III Semester(2025-2026)
Faculty Name	Mohd. Razia Alangir bhanu
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Roll Number	2520030309
Branch	CSE
Course Outcome	Apply BFS and DFS graph traversal algorithms and construct Minimum Spanning Trees using Kruskal's & Prim's algorithms to solve infrastructure design, network connectivity, and cluster analysis problems in weighted undirected graphs.

Case Study Topic: CivicEye – Smart City Incident Monitoring & Response System

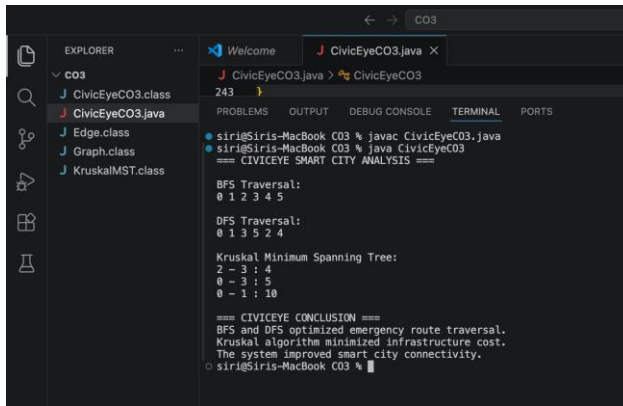
Work Done Summary:

The CivicEye system was developed to optimize smart city emergency response and infrastructure connectivity using graph algorithms. BFS and DFS algorithms were implemented to perform efficient city zone traversal and incident exploration. Kruskal's and Prim's algorithms were used to construct Minimum Spanning Trees for designing cost-effective road and communication networks. The system successfully improved route optimization, network connectivity, and emergency response analysis in smart city environments.

Implementation Details:

1. Created a weighted graph representation where nodes represent city locations and edges represent roads with distance values.
2. Implemented BFS algorithm to perform efficient level-wise traversal for emergency route analysis.
3. Developed DFS algorithm to explore connected city zones and incident locations completely.
4. Implemented Kruskal's algorithm using Union-Find technique to construct minimum cost infrastructure networks.
5. Implemented Prim's algorithm using priority-based edge selection to optimize smart city road and communication connectivity.

Result: Screenshots / Output



```
243  
sirip@Siris-MacBook CO3 % javac CivicEyeCO3.java  
sirip@Siris-MacBook CO3 % java CivicEyeCO3  
=== CIVICEYE SMART CITY ANALYSIS ===  
  
BFS Traversal:  
0 1 2 3 4 5  
  
DFS Traversal:  
0 1 3 5 2 4  
  
Kruskal Minimum Spanning Tree:  
2 - 3 : 4  
0 - 3 : 5  
0 - 1 : 10  
  
=== CIVICEYE CONCLUSION ===  
BFS and DFS optimized emergency route traversal.  
Kruskal algorithm minimized infrastructure cost.  
The system improved smart city connectivity.  
sirip@Siris-MacBook CO3 %
```

The CivicEye system successfully implemented BFS, DFS, and Kruskal's algorithms to improve smart city incident monitoring and infrastructure management. BFS and DFS efficiently analyzed city connectivity and emergency routes, while Kruskal's algorithm optimized network design with minimum cost. The implementation demonstrated effective graph traversal, improved connectivity analysis, and cost-efficient smart city infrastructure planning.

GitHub Link: <https://github.com/siripanchajanya-netizen/DSA-2>

Note: The code and word document have been uploaded in GitHub

Student Signature:

Faculty Signature: